SYLLABUS-ALIGNED PROBLEM SET

Problem Set 02

Inverse Matrices and Linear Systems

Show your calculations and reasoning. Write every solution by hand on paper. Upload a clear photograph of each handwritten solution to its matching field in Microsoft Forms. Check your work before submitting.

Problem 1

Find A^{-1} for the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

and verify $AA^{-1} = I$.

Problem 2

Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Show the row operations or solve $AX = I$.

Problem 3

Solve

$$\begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$$

using Cramer's rule. Check the condition for using the rule.

Problem 4

Solve the system

$$\begin{cases} x + y + z = 4 \\ 2x - y + z = 8 \\ x + 2y - z = -3 \end{cases}$$

by Gaussian elimination. Then independently compute z using Cramer's rule on the 3×3 coefficient matrix.

Problem 5

Use the inverse-matrix method to solve

$$\begin{cases} 3x + y = 11 \\ 2x + y = 8 \end{cases}.$$

Compare your answer with direct subtraction.

Problem 6

Classify each system and give its solution set when it exists:

(a) $\begin{cases} x + 2y = 3 \\ 2x + 4y = 6 \end{cases}$

(b) $\begin{cases} x + 2y = 3 \\ 2x + 4y = 7 \end{cases}$.

Problem 7

Discuss the number of solutions of

$$\begin{cases} x + y = 2 \\ x + ky = k + 1 \end{cases}$$

for every real k, and solve whenever possible.

Problem 8

Find all solutions of the homogeneous system

$$\begin{cases} x + 2y - z = 0 \\ 2x + 4y - 2z = 0 \end{cases}.$$

Express the solution as a linear combination of two vectors.

Problem 9

Solve the rectangular system

$$\begin{cases} x + y = 3 \\ 2x - y = 0 \\ 3x = 3 \end{cases}.$$

Explain why three equations in two unknowns need not be inconsistent.

Problem 10

A workshop sells 30 tickets. A standard ticket costs 12 currency units and a student ticket costs 8. Revenue is 304. Form a linear system, solve it, and check that the answer is meaningful.